

第八章习题参考答案

8.1

$$G = \frac{4\pi}{\Omega_B} = \frac{4\pi}{\frac{\pi}{4}\theta_B^2} = \frac{16}{\theta_B^2}$$

$$\theta_B = 1^\circ = \frac{\pi}{180} \times 1 \text{ rad}, \quad \theta_B = 3^\circ = \frac{\pi}{180} \times 3 \text{ rad}, \quad \theta_B = 5^\circ = \frac{\pi}{180} \times 5 \text{ rad}$$

$$G|_{\theta_B=1^\circ} = 10 \lg \frac{16}{\left(\frac{\pi}{180}\right)^2} = 10 \lg 52578 = 47.2 \text{ dB}$$

$$G|_{\theta_B=3^\circ} = 10 \lg \frac{16}{\left(\frac{\pi}{180} \times 3\right)^2} = 37.7 \text{ dB}$$

$$G|_{\theta_B=5^\circ} = 10 \lg \frac{16}{\left(\frac{\pi}{180} \times 5\right)^2} = 33.2 \text{ dB}$$

8.2

在圆球坐标系中有

$$A_r = A_z \cos \theta$$

$$A_\theta = -A_z \sin \theta$$

$$A_\varphi = 0$$

由 $\mathbf{H} = \frac{1}{\mu} \nabla \times \mathbf{A}$, 得

$$\begin{aligned} \mathbf{H} &= \frac{1}{\mu_0} \nabla \times \mathbf{A} = \frac{1}{\mu_0 r^2 \sin \theta} \begin{vmatrix} \mathbf{r}_0 & r\boldsymbol{\theta}_0 & r \sin \theta \boldsymbol{\varphi}_0 \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \varphi} \\ A_r & rA_\theta & r \sin \theta A_\varphi \end{vmatrix} \\ &= \frac{1}{\mu_0 r^2 \sin \theta} \begin{vmatrix} \mathbf{r}_0 & r\boldsymbol{\theta}_0 & r \sin \theta \boldsymbol{\varphi}_0 \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \varphi} \\ A_z \cos \theta & -A_z r \sin \theta & 0 \end{vmatrix} \\ &= -\frac{1}{\mu_0 r} \left(r \sin \theta \frac{\partial}{\partial r} A_z + \frac{\partial}{\partial \theta} A_z \cos \theta \right) \boldsymbol{\varphi}_0 \end{aligned}$$

$$A_z = \frac{\mu_0 I_0}{2\pi k r} \sin \frac{kl}{2} \frac{\cos(\frac{kl}{2} \cos \theta)}{\sin^2 \theta} e^{-jkr}$$

只取 $1/r$ 项, 则:

$$\mathbf{H} \approx -\frac{1}{\mu_0} \sin \theta \frac{\partial A_z}{\partial r} \boldsymbol{\Phi}_0 \approx j \frac{I_0}{2\pi r} \sin \frac{kl}{2} \frac{\cos(\frac{kl}{2} \cos \theta)}{\sin \theta} e^{-jkr} \boldsymbol{\Phi}_0$$

$$l = \lambda/2$$

$$\mathbf{H} = j \frac{I_0}{2\pi r} \frac{\cos(\frac{\pi}{2} \cos \theta)}{\sin \theta} e^{-jkr} \boldsymbol{\Phi}_0$$

$$\mathbf{E} = \frac{1}{j\omega\epsilon} \nabla \times \mathbf{H} = \frac{1}{j\omega\epsilon} \frac{1}{r^2 \sin \theta} \begin{vmatrix} \mathbf{r}_0 & r\boldsymbol{\theta}_0 & r \sin \theta \boldsymbol{\Phi}_0 \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \varphi} \\ 0 & 0 & r \sin \theta H_\varphi \end{vmatrix}$$

$$\approx -\frac{1}{j\omega\epsilon} \frac{\partial}{\partial r} H_\varphi \boldsymbol{\theta}_0 \quad (\text{只取 } 1/r \text{ 项})$$

$$\approx \frac{jk}{\omega\epsilon} \frac{I_0}{2\pi r} \frac{\cos(\frac{\pi}{2} \cos \theta)}{\sin \theta} e^{-jkr} \boldsymbol{\Phi}_0$$

$$= j \frac{\eta_0 I_0}{2\pi r} \frac{\cos(\frac{\pi}{2} \cos \theta)}{\sin \theta} e^{-jkr} \boldsymbol{\Phi}_0$$

$$8.3 \quad G_D = 10^{\frac{45}{10}} = 3.16 \times 10^4$$

$$\langle \mathbf{S}_r \rangle = G_D \frac{P}{4\pi r^2} = 3.16 \times 10^4 \frac{10 \times 10^3}{4\pi \times (50 \times 10^3)^2} = 10.1 \text{ mW/m}^2$$

$$8.4 \quad \langle \mathbf{S}_r \rangle = G_D \frac{P}{4\pi r^2} = 5 * \frac{200}{4\pi \times (10 \times 10^3)^2} = 7.96 * 10^{-7} \text{ W/m}^2$$

$$\langle \mathbf{S}_r \rangle = \frac{1}{2} \frac{|\mathbf{E}|^2}{\eta} \Rightarrow |\mathbf{E}|$$

场强相同，该点处<S>相同，功率不变，r 增加为原来的 2 倍，G 增加为原来的四倍=20

$$8.5. \quad \langle \mathbf{S}_r \rangle = \frac{1}{2} \frac{|\mathbf{E}|^2}{\eta}, \quad \text{各向同性天线, } P = \langle \mathbf{S}_r \rangle \times 4\pi r^2; \quad \text{短振子天线, } P = \langle \mathbf{S}_r \rangle \times \frac{4\pi r^2}{G}$$

$$8.7 \quad P_r = \left(\frac{\lambda}{4\pi r}\right)^2 P_t G_t G_r, r = 500 \text{ km}, P_t = 1 \text{ kW}, \lambda = c/f = 1.5 \text{ m},$$

$\therefore \theta = 45^\circ, G_t = G_r = 0.75 \Rightarrow P_r$ ，这里半波偶极子天线的增益近似为电偶极子天线的增益

了。正确的要用增益公式来计算一下半波振子 45 度时的增益值，代入上面的传输方程。

$$8.8 \quad 1/4 \text{ 波长单极天线等效为半波振子天线, } G=1.64, \langle \mathbf{S}_r \rangle = \frac{1}{2} \frac{|E|^2}{\eta}, P = \langle \mathbf{S}_r \rangle \times \frac{4\pi r^2}{G},$$

因为单极天线的输入阻抗为阵子天线的 1/2, 所以功率为等效天线的 1/2

8.9 该天线由于架在地面上, 由镜像原理等效为相距为 2h 的两个振子的天线阵, 且相位相差 180 度。最大辐射仰角=30°, 所以即当 $\varphi=30^\circ$ 时, 阵因子 F 最大即可。

$$8.13 \quad N = 40, \quad d = \lambda/2, \quad kd = \pi, \quad \psi = 0, \quad \omega = Nd = 20\lambda$$

$$|F(\varphi)| = \frac{1}{N} \left| \frac{\sin \frac{Nkd \sin \varphi}{2}}{\sin \frac{kd \sin \varphi}{2}} \right| = \frac{1}{40} \left| \frac{\sin(20\pi \sin \varphi)}{\sin \frac{\pi}{2} \sin \varphi} \right|$$

$$\text{因为当 } ws' = \pm 0.443, \quad \text{sinc } ws' = 0.707. \quad (ws')_{B/2} = \frac{w}{\lambda} \sin \frac{\varphi_B}{2} = \frac{Nd}{\lambda} \sin \frac{\varphi_B}{2} = 20 \sin \frac{\varphi_B}{2}$$

所以波束宽度为:

$$\frac{20\lambda}{\lambda} \sin \frac{\varphi_B}{2} = 0.443$$

$$\sin \frac{\varphi_B}{2} = 0.02215$$

$$\frac{\varphi_B}{2} = 1.2692^\circ$$

$$\varphi_B = 2.5384^\circ$$

$$\text{第一个零点位置时 } \frac{20\lambda}{\lambda} \sin \frac{\varphi_0}{2} = 1, \quad \sin \frac{\varphi_0}{2} = 0.05, \quad \frac{\varphi_0}{2} = 2.866^\circ, \quad \varphi_0 = 5.732^\circ$$

$$8.15 \quad G_D = 10^{\frac{45}{10}} = 10^{4.5} = 3.16 \times 10^4$$

$$(1) \quad f = 500 \text{ MHz}, \quad A_e = \frac{G_D \lambda^2}{4\pi} = 905.73 \text{ m}^2$$

$$A' = \frac{A_e}{60\%} = \frac{5}{3} A_e = 1509.55 \text{ m}^2$$

$$d = \sqrt{\frac{4A'}{\pi}} = 43.9 \text{ m}$$

$$(2) \quad f = 40 \text{ GHz}, \quad A_e = \frac{G_D \lambda^2}{4\pi} = 0.142 \text{ m}^2$$

$$A' = \frac{A_e}{60\%} = 0.236 \text{ m}^2$$

$$d = \sqrt{\frac{4A'}{\pi}} = 0.548m^2$$

8.16

电基本振子远区的辐射电场:

$$\mathbf{E}_e = \mathbf{0}_0 \sqrt{\frac{\mu}{\varepsilon}} \frac{jkI_1 l_1 e^{-jkr}}{4\pi r} \sin \theta$$

磁基本振子远区的辐射电场:

$$\mathbf{E}_m = \mathbf{\Phi}_0 \sqrt{\frac{\mu}{\varepsilon}} \frac{I_2 S_2 k^2 e^{-jkr}}{4\pi r} \sin \theta$$

总辐射电场:

$$\begin{aligned} \mathbf{E} &= \mathbf{0}_0 \sqrt{\frac{\mu}{\varepsilon}} \frac{jkI_1 l_1 e^{-jkr}}{4\pi r} \sin \theta + \mathbf{\Phi}_0 \sqrt{\frac{\mu}{\varepsilon}} \frac{I_2 S_2 k^2 e^{-jkr}}{4\pi r} \sin \theta \\ &= (\mathbf{0}_0 jI_1 l_1 + \mathbf{\Phi}_0 kI_2 S_2) \sqrt{\frac{\mu}{\varepsilon}} \frac{ke^{-jkr}}{4\pi r} \sin \theta \end{aligned}$$

8.17

由上题可知, 远区相叠加为一圆极化电磁波, 则 $I_1 l_1 = \pm kI_2 S_2$

$$p = \frac{I_1 l_1}{j\omega}, \quad m = I_2 S_2$$

$$\text{所以, } j\omega p = \pm km, \quad \frac{p}{m} = \pm j\sqrt{\varepsilon\mu}$$

$$8.22 \quad G_D = 10^{\frac{45}{10}} = 3.6 \times 10^4$$

$$P_{R \min} = -115 \text{ dBm} = 10^{-115/10} \times 1(\text{mW}) = 3.16 \times 10^{-12} \text{ mW},$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{5 \times 10^9} = 0.06 \text{ m}, \sigma = 1 \text{ m}^2, P_T = 1 \text{ MW}$$

$$R_{\max} = \left[\frac{P_T G^2 \lambda^2 \sigma}{(4\pi)^3 P_{R \min}} \right]^{1/4} = 0.87 \times 10^6 \text{ m}$$